Evaluation of the New NAIC Economic Scenario Generator

PSTAT 296B

Team 3: Hu, Zexing; Tang, Yuting; Xu, Ao

Instructor: Professor Hal Pedersen

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I. Objective

The aim of this project is to evaluate the performance of two ESG models, which are the AIRG model and NAIC model. First step taken is to analyze historical data to gain insight into market behavior and form capital market assumptions, for which the result is delivered in the interim report. Based on the assumptions, acceptance criteria are then established for comparing the strengths and weaknesses of the two models.

II. Background

2.1 NAIC and Risk-based Capital Requirements

The NAIC is an entity that provides expertise, data, and analysis for insurance commissioners to regulate the industry and protect consumers effectively (NAIC - Our Story, n.d.). In 2017, the American Academy of Actuaries(AAA) stopped maintaining the ESG. Then, the NAIC Life and Annuity Task Force (LATF) authorized its vendor, Conning, to provide, maintain, and support the generator (NAIC Economic Scenario Generator Project - Overview, n.d.). In 2022, NAIC released a new ESG model (NAIC ESG) and is considering replacing the previously used Academy's Interest Rate Generator (AIRG).

Risk-based Capital Requirements (RBC) are imposed by the regulators to ensure that insurance companies can fulfill their financial obligations to policyholders (Risk-based capital, n.d.). The RBC requirement is a statutory minimum level of capital based on the company's size and the inherent riskiness of its financial assets and operations (Risk-based capital, n.d.).

2.2 Economic Scenario Generator

ESG is used to produce real-world interest and equity scenarios that can be prescribed for use in the calculations of Life and Annuity reserves and capital in the United States under the NAIC RBC requirements (NAIC Economic Scenario Generator Project - Overview, n.d.).

There are three component models in ESG, which are:

1. Treasury Interest Rate Model
2. Bond Fund Model
3. Equity Return Model

Among the three components, the Treasury Interest Rate Model is the core model of the generator. It is a standard finance Cox-Ingersoll-Ross 3-factor interest rate model that is used to simulate future treasury yield rates. The Bond Fund Model deals with the extra complexities arising from migration risk and credit spread dynamics that lead to fluctuations in bond prices (Pedersen et al., Economic Scenario Generators - A Practical Guide 2016). And the Equity Model is used to provide returns for representative equity funds offered in US Life Insurance products (NAIC Economic Scenario Generator Project - ESG Models, n.d.).

As depicted in the screenshot of the AIRG model's user interface page, users are prompted to enter initial yield curve values and select the number of years to project, the number of simulated scenarios, time steps, and output rates.

If the user selects the displayed options, the generator will produce a 1000-by-40 matrix with the number of rows representing the number of simulated paths and the number of columns representing the quarterly bond rates for the next ten years.

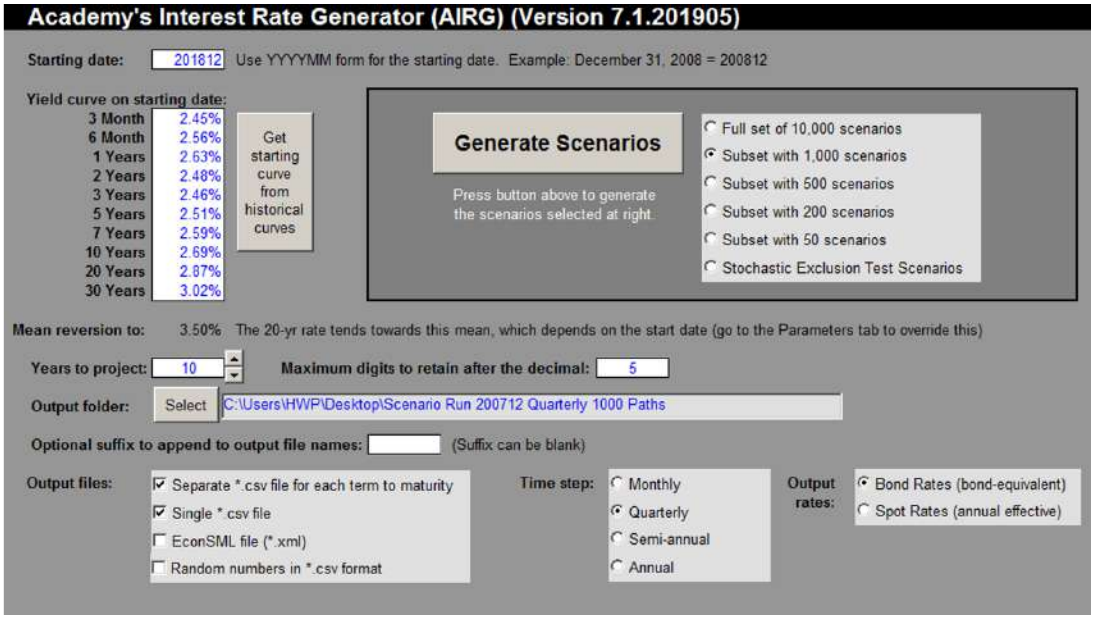


Figure 2.3: Screenshot of the User Interaction Page of the AIRG Scenario Generator

2.3 Capital Market Assumption

Year		Inflation	Fed	1m	5Yr	20Yr
	Today	7.70%	3.83% (Actual)	4.022% (Actual)	3.89% (Actual)	3.98% (Actual)
1	End of 2023	4.00%	4.30%	3.80%	4.80%	5.50%
2	End of 2024	3.50%	3.80%	3.40%	4.40%	5.20%
3	End of 2025	3.00%	3.40%	3.00%	4.10%	4.90%
5	End of 2027	2.50%	2.90%	2.60%	3.70%	4.70%
20	End of 2042	2.50%	2.90%	2.60%	3.70%	4.70%

Table 2.3: Capital Market Assumption Summary Table from 296A Interim Report

The numbers in the table above are concluded from quantitative analyses using standard econometric modeling techniques as well as expert judgments derived from research on historical periods. Please refer to the interim report for more information on how these numbers are derived.

Inflation is expected to go back down to around 3 percent by the end of 2025, and to around 2.5% by the end of 2027. This is based on the previous analysis that the Federal Reserve will manage to track inflation back to its target of 2%. However, on average, the Federal Reserve missed its target by about 0.5%, making a long-term inflation rate estimate of 2.5% plausible.

For the other types of securities, in the long run, the 1-Month Treasury Bill Yield is expected to be around 2.6%, the 5-Year Bond Yield is expected to be around 3.7%, and the 20-Year Bond Yield is expected to be around 4.7%.

The table's numbers will be used to establish a set of acceptance criteria for evaluating the new NAIC ESG. If the generator is functioning properly, the model's average simulated output should be close to the numbers presented above.

III. Datasets

3.1 SBBI Data

The primary dataset is known as the SBBI dataset. SBBI stands for Stocks, Bonds, Bills, and Inflation. This dataset mostly includes Total Return and Yield, such as Large Capital Total Return, Long-Term Government Total Return, Intermediate-Term Government Total Return, Long-Term Corporate Total Return, 30-Day T-Bill Total Return, Long-Term Government Yield, and Intermediate-Term Government Yield. Yield is the sum that an investment earns over a given period of time. It is the revenue that a fund disburses on a monthly or quarterly basis. Total Return is the amount that an investment makes or loses over time. Interest payments made by the bonds held in the fund have an impact on total return.

3.2 FED Data

For obtaining capital market assumptions, the Federal Reserve's economic data and world financial data are useful. It primarily consists of the FED Funds Rate, the One-Month T-Bill Yield, and the Year-over-Year CPI Inflation Rate.

3.3 AIRG Simulated Data

Recently, the AIRG model has been made publicly available. In light of this development, we decided to test the model's accuracy by feeding it input data from 1950 to 2021 and predicting the data for 2022. For our analysis, we used the data results generated by the AIRG model.

3.4 NAIC Simulated Data

As the Conning company did not release their model to the public, it’s opted to use the test data (NAIC ESG Field Test #1a: Scenario Sets and Statistics: 1000 Path Subset 1a Conning GFF Baseline Equity 123121) provided by the company to conduct our analysis.

IV. Methodology

To evaluate the effectiveness and accuracy of the two economic scenario generators, we will use statistical analysis to compare the performance of the three primary components of the generators, which are the treasury interest rate model, equity return model, and bond return model. By examining these components individually, we can gain a deeper understanding of the strengths and weaknesses of each generator, and identify potential areas for improvement.

4.1 Treasury Interest Rate Model

There are two approaches used to analyze the rate level produced by the simulators. The first method involves comparing the Percentiles Exponentially Weighted (PEWs) of historical and simulated data, while the second involves using the backtesting method.

4.1.1 Exponentially Weighted Percentiles

This methodology is adopted from a slidedeck made by the AIRG ESG Work Group. The underlying idea is to give greater importance to the most recent data and less importance to older data when comparing the distribution of data. We deviated slightly from the design proposed by the ESG work group by using a different half-life value for our calculations. Rather than using the ESG work group's 15-year half-life, we opted for a longer half-life of 48.5 years. Our decision was based on the fact that we are working with a dataset that spans a longer time period than the ESG group's dataset. To give more weight to recent data, we selected a longer half-life value. The formula used to calculate the weight is given as $w_i = (1 - \alpha)^{n-i}$. In this formula, n represents the total number of months in the dataset, and i represents the i_{th} month in the dataset. For instance, if the ARIG simulator is set to generate 30 years of monthly data for the 20-year US Treasury yield, then n would be calculated as 30 multiplied by 12, which equals 360. The first simulated monthly data point would correspond to $i = 1$, while the most recent data point would correspond to $i = 360$.

The value of α was determined based on the SBBI data, which covers the time period from January 1, 1926 to December 31, 2022, and includes a total of 1164 monthly data points. To calculate the value, we used the following formula: $\frac{1}{2} = (1 - \alpha)^{\frac{1164}{2}}$, which resulted in $\alpha = 0.00119$. Once we have established the α value, we will proceed to calculate the corresponding w_i value for each data point in our historical data as well as the simulated data generated by both models. These weights will then be normalized to calculate the exponential weighted percentile. As our datasets have varying dimensions, the weight calculation process differs slightly. Please see the pseudocode provided below, and refer to the attached appendix for the complete code and additional details.

For Historical Data:

$$w_i = (1-\alpha)^{n*12-i}$$

$$nw_i = \frac{w_i}{\sum_i^{n*12} w_i}$$

$$cbind(data, N\bar{W})$$

For Simulated Data:

$$W[1,] = (w_1, w_2, \ldots w_N)$$
$$rep(W[1,]) \times 1000$$
$$NW[i,j] = \frac{W[i,j]}{\sum W}$$
$$NW \sim (1000 \times N)Matrix; \sum NW = 1$$
$$NAIC \sim (1000 \times 1200)Matrix; AIRG \sim (1000 \times 828)Matrix;$$
$$cbind(unlist(NAIC), unlist(NW))$$

Figure 4.1.1: Pseudocode for Calculating the Exponentially Weighted Percentiles

We have calculated the exponential weighted percentiles for the yields of two US Treasury bonds with different maturities: the 3-month and 20-year bonds. The reason for examining both short-term and long-term bonds is because their yields can affect the overall shape of the yield curve, which is a key indicator of the performance of the bond market. Specifically, the shape of the yield curve can provide valuable insights into the market's expectations for future economic growth and inflation. The exponential weighted percentiles will allow us to determine how the

yields of these bonds have varied over time, and compare these trends to historical data to evaluate the accuracy and effectiveness of our models.

4.1.2 Backtesting

Backtesting, utilizing historical data, is the evaluation process we used to assess the reliability and validity of the two predictive models. It involves creating a simulated environment that accurately reflects the past conditions and enables a detailed analysis of performance. This method involves a thorough comparison of simulated data against historical data for US 3-month Treasury Bills and US 20-year Treasury Bills at the reference year of 2022. We first calculate the minimum and maximum values, as well as various percentiles (10%, 20%, 50%, 80%, and 90%) , for simulated data produced by the two models. It then can be compared these values with the corresponding percentiles in the historical data. Through careful examination of how well the range of simulated paths aligns with the historical data, it can be assessed the effectiveness of the two ESG models in accurately representing the historical data.

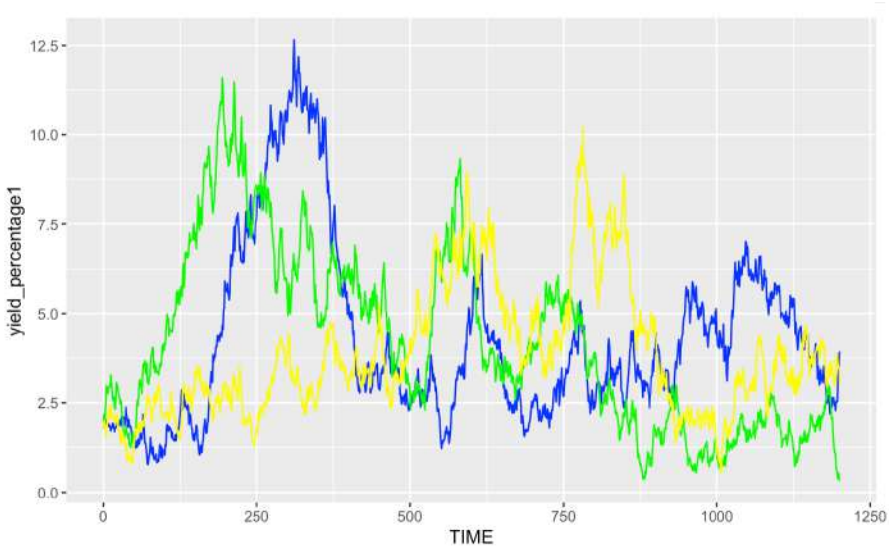


Figure 4.1.2.1: Screenshot of the 3 Example Paths of the Simulated Data

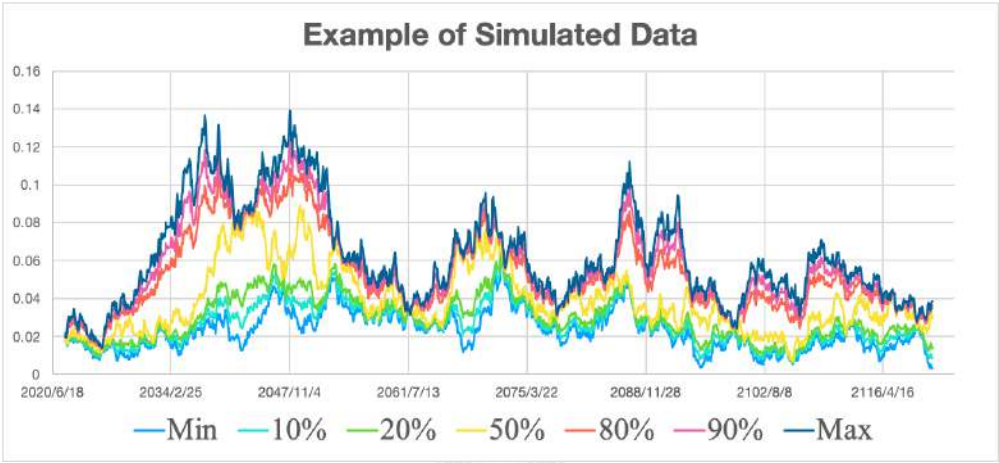


Figure 4.1.2.2: Screenshot of the Example Paths of the Simulated Data

The given information pertains to a simulation study that involves 1,000 simulated pathways for each model. The figure 4.1 displays three randomly selected sample paths out of the total 1,000 simulated paths. To analyze the simulated pathways, statistical measures such as minimum and maximum values, as well as various percentiles (e.g., 10%, 20%, 50%, 80%, and 90%) at each time point like Figure 4.2, can be determined using Excel. This process is applied to all 1,000 simulated pathways to obtain a comprehensive understanding of the simulation study.

4.2 Equity Return Model

4.2.1 Equity Risk Premium

As discussed above, the Equity Model in the ESG is used to provide returns for representative equity funds offered in US Life Insurance products. To evaluate the effectiveness of this component, we will compute the equity risk premium statistics of the historical data and simulated data and use them for comparison. Equity risk premium is defined as $Equity\ Risk\ Premium\ = \frac{Large\ Capital\ Stock\ Total\ Return}{Short\ Term\ US\ Treasury\ Bond\ Total\ Return} - 1$. This is used to measure the reward for risk investors received by comparing the return on equities(typically stocks) with the return from risk free investments, which usually refer to short-term US Treasury Bonds. We chose to use the 1-month US Treasury bond as our benchmark for calculating the equity risk premium because it is a government security with a very short-term maturity, is free of default risk, and has a fixed-income. Unfortunately, the AIRG simulator does not generate simulated data for the 1-month US Treasury bond, so we opted to use the 3-month US Treasury Bond to calculate the equity risk premium for the simulated data generated by the AIRG model.

4.3 Bond Return Model

4.3.1 Bond Risk Premium

As discussed earlier, the bond return model deals with the extra complexities arising from migration risk and credit spread dynamics that lead to fluctuations in bond prices. As with the previous section, we will calculate risk premium statistics to evaluate the effectiveness of the bond fund model of the two simulators. Specifically, we will compare the results generated by the simulators with historical values to assess their performance. In doing so, we will be able to determine whether the models are able to capture the complexities of the bond market. By comparing the simulated results with historical values, we can gauge the accuracy of the models and assess their suitability for use in investment decision-making.

The formula is defined as $Bond\ Risk\ Premium\ = \frac{Bond\ Total\ Return}{Short\ Term\ US\ Treasury\ Bond\ Total\ Return} - 1$. And this is used to measure the reward for risk investors received by comparing the return on bonds (intermediate/ long term) with the return from risk free investments (short term government bonds). Similarly, we will use 1 month and 3 month US Treasury Bonds as our benchmark for the same reasons as we discussed in the previous section.

V. Analytic Results

5.1 Exponentially Weighted Percentiles

5.1.1 3-Month US Treasury Yield

The colored marks in the simulation results indicate significant differences between the simulated data and historical data. Green marks signify that the simulated data is significantly lower than the historical data, while red marks indicate that the simulated data is significantly higher than the historical data.

Overall, both models perform well in simulating low rates and extreme values, which is consistent with the behavior of the market that was analyzed in the previous quarter. However, when examining the tail distribution of the two models (i.e., the minimum and maximum values), there are noticeable differences. Specifically, the data simulated by the NAIC model has a heavier tail and covers a wider range, suggesting that it is more volatile and unstable compared to the AIRG model.

	Lower than Hist	Higher than Hist	
US Treasury Yield - 3 Mo. (PEWs)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-0.01%	0.01%	-1.15%
1%	0.01%	0.01%	-0.64%
5%	0.04%	0.39%	-0.35%
15%	0.14%	0.82%	-0.03%
30%	1.56%	1.23%	0.30%
50%	3.55%	1.73%	1.94%
70%	5.11%	2.34%	4.37%
85%	6.97%	3.10%	7.13%
95%	9.18%	4.33%	11.02%
99%	14.22%	6.34%	16.33%
Max	15.52%	16.78%	38.09%

Table 5.1.1: Exponentially Weighted Percentiles of 3-Month US Treasury Yield

5.1.2 20-Year US Treasury Yield

The results in this section are consistent with the findings from the previous section. The NAIC model continues to produce simulated data that covers a wider range than the AIRG model. However, the AIRG model performs better in producing low rates that are similar to the historical scenario.

Overall, the results suggest that the two models have their respective strengths and weaknesses in simulating data. The wider range produced by the NAIC model may be useful for capturing extreme scenarios or tail risk, but it may also indicate higher volatility and instability. On the other hand, the better performance of the AIRG model in producing low rates closer to historical data may be more suitable for modeling stable and predictable scenarios.

	Lower than Hist	Higher than Hist	
US Treasury Yield - 20 Yr (PEWs)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	0.97%	0.92%	0.14%
1%	1.17%	1.44%	0.38%
5%	2.00%	1.81%	1.00%
15%	2.46%	2.18%	1.81%
30%	3.03%	2.54%	2.72%
50%	4.38%	2.95%	3.99%
70%	6.23%	3.45%	5.60%
85%	8.10%	4.10%	7.55%
95%	10.95%	5.16%	10.45%
99%	13.48%	6.88%	14.79%
Max	14.82%	16.86%	33.52%

Table 5.1.1.2: Exponentially Weighted Percentiles of 20-Year US Treasury Yield

5.1.3 Steady State

To further analyze the performance of the two models, we also examine the distribution of simulated data and historical data during the steady state. The distribution of data during the steady state can be valuable in providing us with useful information. This is because, as the simulation process runs for a longer period of time, the starting point becomes less significant. Consequently, the steady state data distribution should be able tp represent the overall characteristics of the data set.

To produce the histograms below, we took the most recent 10 years of data from each dataset. The histograms in the first row represent results from the 3 month US Treasury Yield datasets, while the histograms in the second row represent results from the 20 year US Treasury Yield datasets. Examining the 3 month data, the histograms of our simulated data support the findings from the previous sections. For the AIRG model, most of the simulated data falls within the range of 0 to 2.5%, while for the NAIC model, the density of low rates is higher than that of higher rates. These results indicate that both models are more likely to simulate low rates than high rates. Additionally, the heavy tail on the right side of the histograms shows that both models are capable of simulating extreme values that represent unusual market behavior.

For the 20 year data, we observe similar patterns as in the 3 month data, and the same conclusions can be drawn. Therefore, we do not need to go into further detail here. These results provide further evidence of the strengths and weaknesses of the two models in simulating data and can help inform risk management decision-making.

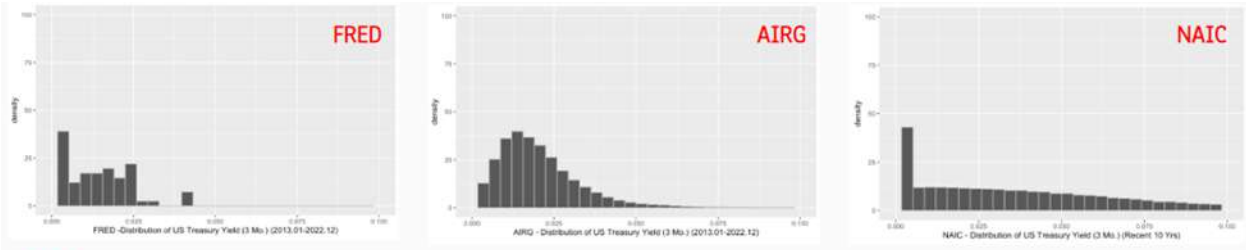


Figure 5.1.1.3.1: Distributions of Exponentially Weighted Percentiles of 3-Month US Treasury Yield

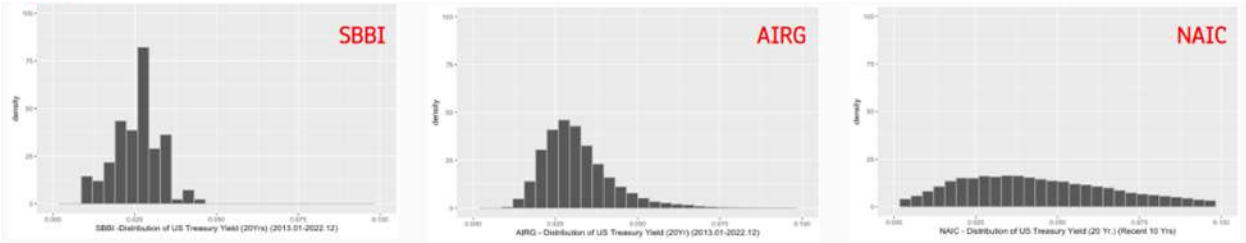


Figure 5.1.1.3.2: Distributions of Exponentially Weighted Percentiles of 20-Year US Treasury Yield

5.2 Backtesting

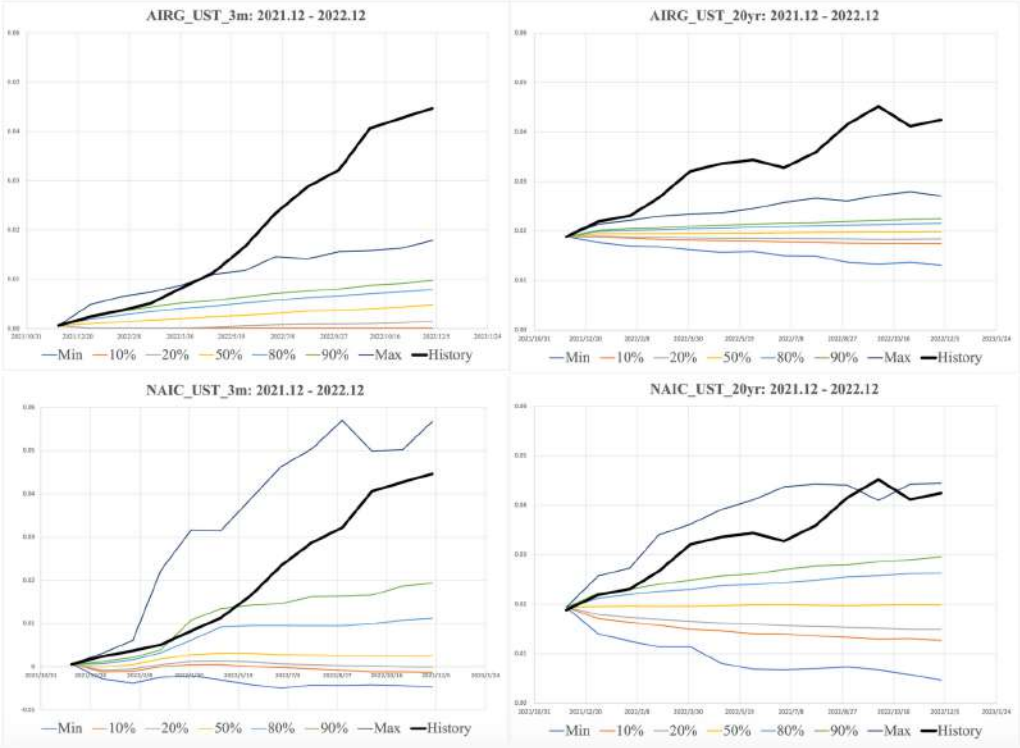


Figure 5.1.2:Backtesting Results for 3 Month US Treasury Yields (Left) and 20 Year US Treasury Yields (Right)

Throughout the year 2022, the recorded data for the US 3-month Treasury Yields and US 20-year Treasury Yields demonstrated a significant increase. As a result, these figures were beyond the expected range for the majority of the simulated paths in the AIRG model. However, the NAIC model is able to generate results that reflect the fluctuation of the historical data.

The 3-month rate and 20-year rate are crucial benchmarks in the financial industry, as they have a substantial influence on the interest rates for various lending products, such as mortgages and corporate bonds. The precise prediction of these rates can significantly affect businesses and households nationwide. The forecast model of the NAIC, by exhibiting an aptitude for accurately projecting future trends, can furnish market participants with valuable insights to make informed decisions regarding their investments and finances.

Based on the outcomes of the backtesting, the NAIC model is capable of predicting a broader range of data that encompasses the range of historical data. On the other hand, the AIRG model generates more stable data but does not produce data that accurately reflect certain unusual market behavior.

5.3 Equity Risk Premium

Upon examination of the table, it is apparent that the distributions within the 1st and 99th percentile of the simulated data from both models are highly similar to the distribution of the historical data. However, the maximum values of the simulated data from both models indicate that the distribution has a heavy tail to the right. This suggests that the models occasionally produce outliers that users should be aware of. When forming an opinion using the simulated data, users should consider excluding these outliers from their analysis.

	Lower than Hist	Higher than Hist	
Equity Risk Premium (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-44%	-61%	-62%
1%	-38%	-30%	-29%
5%	-24%	-19%	-18%
10%	-14%	-13%	-12%
25%	-3%	-4%	-3%
50%	9%	7%	7%
75%	22%	16%	15%
90%	31%	26%	25%
95%	37%	33%	33%
99%	51%	48%	51%
Max	54%	160%	156%

Table 5.2: Distribution of Equity Risk Premium

5.4 Bond Risk Premium

5.4.1 Intermediate and Long-term Government Bond

We have conducted an analysis to calculate the bond risk premium for both intermediate and

long-term government bonds. Please note that we were not able to obtain the bond risk premium results of the AIRG model for long-term government bond because this model does not generate simulated data for that variable.

When we compared the results to our previous analysis on equity risk premium, we found similar patterns. Upon examining the tables, we observed that, within the 5th and 95th percentile, the distributions of our simulated data from the two models are quite similar to the distribution of the historical data. However, similar to our findings in the equity risk premium analysis, the maximum and minimum values of the two models indicate that the distributions of the simulated data from the two models have heavy tails. This suggests that there may be instances where the two models generate data points that are outside the norm and should be considered as outliers. Users should be aware of this while using simulated data to make decisions.

	Lower than Hist	Higher than Hist	
Bond Risk Premium - Intermediate (5YR) (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-9%	-26%	-31%
1%	-7%	-8%	-16%
5%	-5%	-5%	-11%
10%	-3%	-4%	-8%
25%	-2%	-2%	-3%
50%	2%	1%	1%
75%	4%	3%	6%
90%	9%	5%	11%
95%	10%	7%	15%
99%	12%	10%	22%
Max	17%	29%	49%

Table 5.3.1.1: Distribution of Bond Risk Premium (Intermediate Government Bond)

	Lower than Hist	Higher than Hist	
Bond Risk Premium - Long-Term (20YR) (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-15%	NA	-41%
1%	-14%	NA	-24%
5%	-11%	NA	-17%
10%	-8%	NA	-13%
25%	-3%	NA	-6%
50%	3%	NA	2%
75%	7%	NA	9%
90%	15%	NA	17%
95%	22%	NA	23%
99%	27%	NA	34%
Max	27%	NA	76%

Table 5.3.1.2: Distribution of Bond Risk Premium (Long-term Government Bond)

VI. Discussion

Based on the analysis of the exponentially weighted percentiles, we discover that both simulators

can replicate low rates and simulate extreme values, which is consistent with historical market behavior. However, when we look at the tail distribution of the simulated data from both models, we find that the rates produced by the NAIC model are more unstable compared to the AIRG Model.

Based on the outcomes of the backtesting, the NAIC model is capable of predicting a broader range of data that encompasses the range of historical data. On the other hand, the AIRG model generates more stable data but does not produce data that accurately reflect certain unusual market behavior.

Based on the results from equity/bond risk premium, we discover that the majority of the simulated data from both models match the distribution of the historical data. Nevertheless, the minimum and maximum values produced by both models suggest that the distribution of the simulated data has a heavy tail on both sides. This implies that the models may occasionally generate outliers that users should be cautious of. If users intend to utilize the simulated data to make decisions, they should consider disregarding those data points.

In conclusion, we propose that both models have their advantages and disadvantages, and it is difficult to determine which one is better. However, we consider the AIRG Model a better choice for users who want more conservative results compared to the NAIC model.

VIII. Reference

[1] *NAIC RISK-BASED CAPITAL.* (2022, December 6).
Retrieved from https://content.naic.org/cipr-topics/risk-based- capital#:~:text= The%20RBC%20requirement%20is%20a,in%20proportion%20to%20its%20risk

[2] Pasaogullari, M., & Meyer, B. (2010). *Simple Ways to Forecast Inflation: What Works Best?*
FEDERAL RESERVE BANK of CLEVELAND. Retrieved from https://doi.org/10.265 09/frbc-ec-201017

[3] *Why does the Federal Reserve aim for inflation of 2 percent over the longer run?*
BOARD OF GOVERNORS of the FEDERAL RESERVE SYSTEM.(2020). Retrieved from https://www.federalreserve.gov/faqs/economy_14400.htm

[4] Boesler, M., & Bloomberg. (2022, September 21). *The Fed raises interest rates by 75 basis points for the third time this year.*
FORTUNE. Retrieved from https://fortune.com/2022/09/21/fed-raises-interest-rate- 75-ba sis-points/

[5] Kehrberg, J., & Richardson, L. (2022). (rep.). *Interest Rates - Stylized Facts and Acceptance Criteria* (pp. 1–43).
National Association of InsuranceCommissioners (NAIC) Life Actuarial (A) Task Force (LATF).

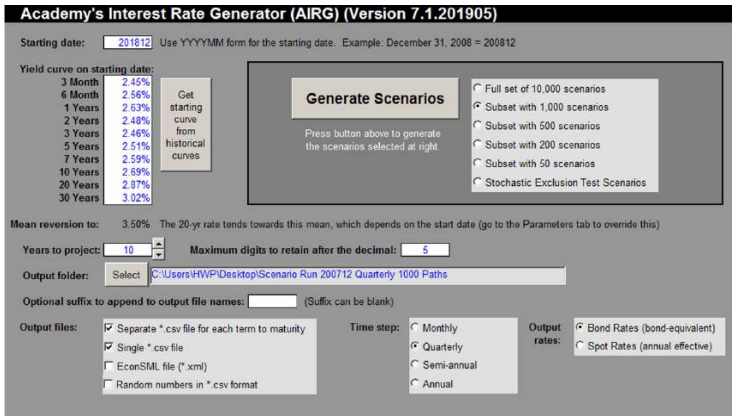
[6] Duff & Phelps. (2016). *2016 SBBI Stocks, Bonds, Bills, and Inflation*. John Wiley & Sons, Inc., Hoboken, New Jersey.

[7] *Scenario Files.* National Association of Insurance Commissioners. (2023).
Retrieved from https://naic.conning.com/scenariofiles

[8] *Economic Scenario Generators.* AMERICAN ACADEMY of ACTUARIES.(2023).
Retrieved from https://www.actuary.org/content/economic-scenario-generators

IX. Appendix

1. Screenshot of the User Interaction Page of the AIRG Scenario Generator



2. Capital Market Assumption Summary Table from 296A Interim Report

Year		Inflation	Fed	1m	5Yr	20Yr
	Today	7.70%	3.83% (Actual)	4.022% (Actual)	3.89% (Actual)	3.98% (Actual)
1	End of 2023	4.00%	4.30%	3.80%	4.80%	5.50%
2	End of 2024	3.50%	3.80%	3.40%	4.40%	5.20%
3	End of 2025	3.00%	3.40%	3.00%	4.10%	4.90%
5	End of 2027	2.50%	2.90%	2.60%	3.70%	4.70%
20	End of 2042	2.50%	2.90%	2.60%	3.70%	4.70%

3. Pseudocode for Calculating the Exponentially Weighted Percentiles

For Historical Data:

$$w_i = (1-\alpha)^{n*12-i}$$

$$nw_i = \frac{w_i}{\sum_i^{n*12} w_i}$$

$$cbind(data, N\bar{W})$$

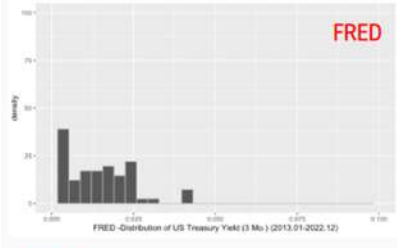
For Simulated Data:

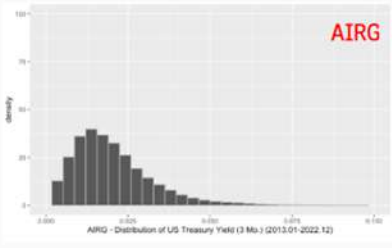
$$W[1,] = (w_1, w_2, \ldots w_N)$$
$$rep(W[1,]) \times 1000$$
$$NW[i,j] = \frac{W[i,j]}{\sum W}$$
$$NW \sim (1000 \times N)Matrix; \sum NW = 1$$
$$NAIC \sim (1000 \times 1200)Matrix; AIRG \sim (1000 \times 828)Matrix;$$
$$cbind(unlist(NAIC), unlist(NW))$$

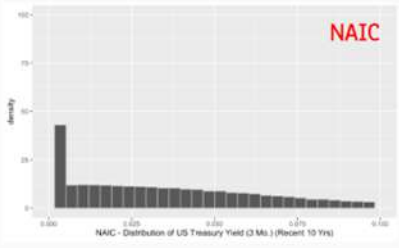
4. Exponentially Weighted Percentiles of 3-Month US Treasury Yield

	Lower than Hist	Higher than Hist	
US Treasury Yield - 3 Mo. (PEWs)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-0.01%	0.01%	-1.15%
1%	0.01%	0.01%	-0.64%
5%	0.04%	0.39%	-0.35%
15%	0.14%	0.82%	-0.03%
30%	1.56%	1.23%	0.30%
50%	3.55%	1.73%	1.94%
70%	5.11%	2.34%	4.37%
85%	6.97%	3.10%	7.13%
95%	9.18%	4.33%	11.02%
99%	14.22%	6.34%	16.33%
Max	15.52%	16.78%	38.09%

5. Distributions of Exponentially Weighted Percentiles of 3-Month US Treasury Yield



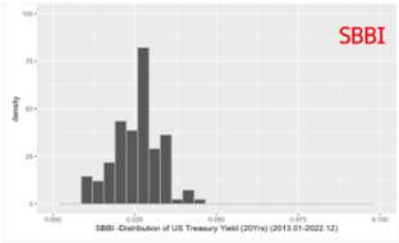


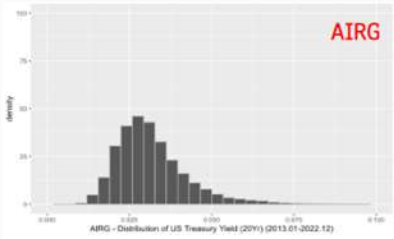


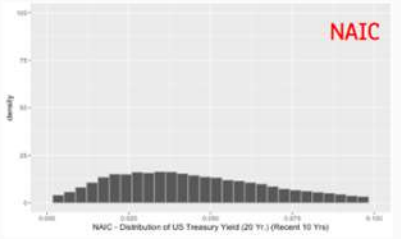
6. Exponentially Weighted Percentiles of 20-Year US Treasury Yield

	Lower than Hist	Higher than Hist	
US Treasury Yield - 20 Yr (PEWs)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	0.97%	0.92%	0.14%
1%	1.17%	1.44%	0.38%
5%	2.00%	1.81%	1.00%
15%	2.46%	2.18%	1.81%
30%	3.03%	2.54%	2.72%
50%	4.38%	2.95%	3.99%
70%	6.23%	3.45%	5.60%
85%	8.10%	4.10%	7.55%
95%	10.95%	5.16%	10.45%
99%	13.48%	6.88%	14.79%
Max	14.82%	16.86%	33.52%

7. Distributions of Exponentially Weighted Percentiles of 20-Year US Treasury Yield







8. Distribution of Equity Risk Premium

	Lower than Hist	Higher than Hist	
Equity Risk Premium (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-44%	-61%	-62%
1%	-38%	-30%	-29%
5%	-24%	-19%	-18%
10%	-14%	-13%	-12%
25%	-3%	-4%	-3%
50%	9%	7%	7%
75%	22%	16%	15%
90%	31%	26%	25%
95%	37%	33%	33%
99%	51%	48%	51%
Max	54%	160%	156%

9. Distribution of Bond Risk Premium (Intermediate Government Bond)

	Lower than Hist	Higher than Hist	
Bond Risk Premium - Intermediate (5YR) (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-9%	-26%	-31%
1%	-7%	-8%	-16%
5%	-5%	-5%	-11%
10%	-3%	-4%	-8%
25%	-2%	-2%	-3%
50%	2%	1%	1%
75%	4%	3%	6%
90%	9%	5%	11%
95%	10%	7%	15%
99%	12%	10%	22%
Max	17%	29%	49%

10. Distribution of Bond Risk Premium (Long-term Government Bond)

	Lower than Hist	Higher than Hist	
Bond Risk Premium - Long-Term (20YR) (Annual)			
Percentile	**Historical Data**	**AIRG**	**NAIC**
Min	-15%	NA	-41%
1%	-14%	NA	-24%
5%	-11%	NA	-17%
10%	-8%	NA	-13%
25%	-3%	NA	-6%
50%	3%	NA	2%
75%	7%	NA	9%
90%	15%	NA	17%
95%	22%	NA	23%
99%	27%	NA	34%
Max	27%	NA	76%

11. Code for backtesting - simulating example paths, calculating percentiles

```{r}
# Figure 4.1.2.1:
# Screenshot of the 3 Example Paths of the Simulated Data
path1 <- UST_20yr[UST_20yr$Scen == 4,]
path1$yield_percentage1 <- path1$US.Treasury.20.Year.Yield...Coupon
t <- path1[,-1]
ti <- t[,-2]
time <- ti[,-2]
df1 <- path1[,-1:-3]
df1
path2 <- UST_20yr[UST_20yr$Scen == 5555,]
path2$yield_percentage2 <- path2$US.Treasury.20.Year.Yield...Coupon
df2 <- path2[,-1:-3]
df2
path3 <- UST_20yr[UST_20yr$Scen == 9974,]
path3$yield_percentage3<- path3$US.Treasury.20.Year.Yield...Coupon
df3 <- path3[,-1:-3]
df3
combine <- cbind(time,df1, df2, df3)
ggplot(combine, aes(x=TIME)) +
  geom_line(aes(y = yield_percentage1), color ="blue") +
  geom_line(aes(y = yield_percentage2), color="green") +
  geom_line(aes(y = yield_percentage3), color="yellow")
```

	44531	44562	44593	44621	44652
min	=MIN(B11:B13)	=MIN(C11:C13)	=MIN(D11:D13)	=MIN(E11:E13)	=MIN(F11:F13)
0.1	=PERCENTILE(B$11:B$13,$A3)	=PERCENTILE(C$11:C$13,$A3)	=PERCENTILE(D$11:D$13,$A3)	=PERCENTILE(E$11:E$13,$A3)	=PERCENTILE(F$11:F$13,$A3)
0.2	=PERCENTILE(B$11:B$13,$A4)	=PERCENTILE(C$11:C$13,$A4)	=PERCENTILE(D$11:D$13,$A4)	=PERCENTILE(E$11:E$13,$A4)	=PERCENTILE(F$11:F$13,$A4)
0.5	=PERCENTILE(B$11:B$13,$A5)	=PERCENTILE(C$11:C$13,$A5)	=PERCENTILE(D$11:D$13,$A5)	=PERCENTILE(E$11:E$13,$A5)	=PERCENTILE(F$11:F$13,$A5)
0.8	=PERCENTILE(B$11:B$13,$A6)	=PERCENTILE(C$11:C$13,$A6)	=PERCENTILE(D$11:D$13,$A6)	=PERCENTILE(E$11:E$13,$A6)	=PERCENTILE(F$11:F$13,$A6)
0.9	=PERCENTILE(B$11:B$13,$A7)	=PERCENTILE(C$11:C$13,$A7)	=PERCENTILE(D$11:D$13,$A7)	=PERCENTILE(E$11:E$13,$A7)	=PERCENTILE(F$11:F$13,$A7)
max	=MAX(B11:B13)	=MAX(C11:C13)	=MAX(D11:D13)	=MAX(E11:E13)	=MAX(F11:F13)

12. Please visit the link provided below to access the full code documentions regarding clearing the NAIC data, calculating PEWS, Equity Risk Premium, and Bond Risk Premium.

https://github.com/zexze/PSTAT-296B-ESG-PROJECT